

An introduction to Causality and Growth Rates

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Today's plan:

- An introduction to causality
- Dealing with Growth Rates
- Review of Slopes

Causality

Economists are often interested in answering the following question: Does an event A cause an event B?

Some examples:

- Does smoking (Event A) increase the chance of developing lung cancer (Event B)?
- Does democracy (Event A) cause growth (Event B)?

In most cases, it is really hard to answer these questions. Seeing that both events happen (correlation) is not sufficient! Why?

- Omitted variable
- Reverse causality

Causality

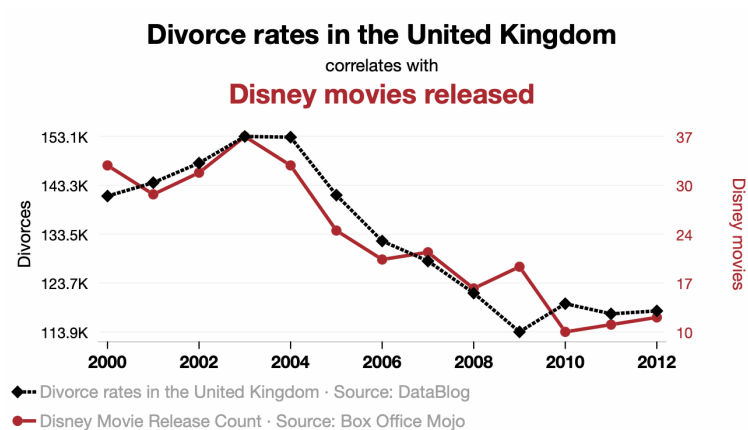
Omitted variable

- A study group observes a strong relationship between two variables: the **number of cigarette lighters** that a person owns and the probability that they will develop **lung cancer** in the future.
- The study group advises the government to discourage people from owning cigarette lighters.
- **Problem:** There is a third variable that is not being considered: **smoking!**
 - People who **smoke** are more likely to **own cigarette lighters** and also more likely to **develop lung cancer**.

Reverse causality:

- One study plots the **number** of **violent crimes** per thousand people in major cities against the **number** of **police officers** per thousand people.
- The study notes the **curve** being **upward sloping** and argues that because police increase rather than decrease the amount of urban violence, law enforcement should be abolished.
- **Problem:** More dangerous cities have more police officers!
 - So rather than police causing crime, **crime causes police**.

Correlation does not imply causation!!!



Source: <https://www.tylervigen.com/spurious-correlations>

Correlation does not imply causation!!!



Funny video

Growth rates

If some variable Y grows at a **constant rate** g , then there is an easy way to find the value of Y at some instant t_1 given its value at some instant t_0 :

$$Y_{t_1} = Y_{t_0} \cdot (1 + g)^{t_1 - t_0}$$

Let's see some examples:

- Suppose the US GDP grows at a constant rate of 2%. How long will it take US to double its GDP?
- US GDP grows at a constant rate of 2% while China's GDP grows at a constant rate of 5%. China's GDP today is 12.6% of US GDP. How long it will take China to surpass the US?

A useful approximation

We can generalize the result we found before.

Suppose US GDP grows at a constant rate of g and you want to find how long it will take for it to be x times the value it is today. Then:

$$t = \frac{\ln(x)}{\ln(1 + g)}$$

Now, for small values of g , we can use the following beautiful approximation:

$$\ln(1 + g) \approx g$$

Which gives us:

$$t = \frac{\ln(x)}{g}$$



Make sure you are using **natural logarithm** when applying this approximation. Otherwise you are going to be very sad with the result.

Slopes

The slope of a line is the “Rise” over the “Run”.

$$\text{slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\Delta Y}{\Delta X} = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Important: We will use the symbol Δ a lot in this class!

- It means **change in** a variable between two scenarios (or two points in time).
- For example, ΔY means **change in** Y , which is equal to $Y_2 - Y_1$.

Consider a line represented by the equation $Y = a + b \cdot X$, then the slope of this line is b .

Why should I care about this?

Slopes

Suppose a variable Y grows at a constant rate of 10% each year and at $t = 0$, $Y_0 = 100$. Using the formula we just learned, Y_t is given by:

$$Y_t = 100 \cdot 1.1^t$$

Taking $\ln()$ on both sides:

$$\ln(Y_t) = \ln(100) + t \ln(1.1)$$

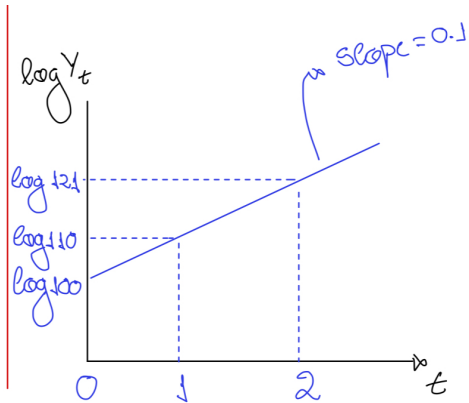
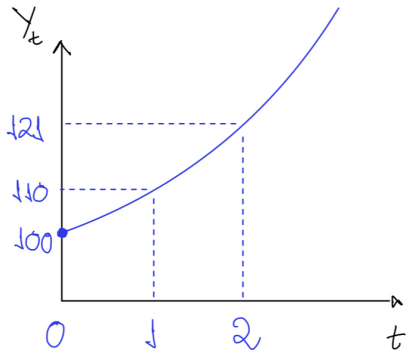
which is a line with slope $\ln(1.1)$.

Using the beautiful approximation we just learned: $\ln(1 + 0.1) \approx 0.1$. Then:

$$\ln(Y_t) = \ln(100) + 0.1 \cdot t$$

The **slope** of the line in the **log-scale** plot is approximately the **growth rate**!

Slopes



Average Growth Rates

Suppose we have a time series on a variable x , represented by: x_1, x_2, x_3 , where the subscript refers to time.

The **rate of growth** of x between $t = 1$ and $t = 2$, g_1 , is:

$$1 + g_1 = \frac{x_2}{x_1}$$

and the **rate of growth** of x between $t = 2$ and $t = 3$, g_2 , is:

$$1 + g_2 = \frac{x_3}{x_2}$$

Note that:

$$x_1 \cdot (1 + g_1) \cdot (1 + g_2) = x_1 \cdot \frac{x_2}{x_1} \cdot \frac{x_3}{x_2} = x_3$$

Average Growth Rates

This gives us a motivation to find the **geometric average rate** of growth of x . Let's call it $\bar{g}$.
 $\bar{g}$ is a **constant growth rate** that satisfies:

$$x_1 \cdot \underbrace{(1 + \bar{g}) \cdot (1 + \bar{g})}_{(1 + \bar{g})^2} = x_3$$

Isolating $\bar{g}$:

$$\bar{g} = \left(\frac{x_3}{x_1} \right)^{1/2} - 1$$

What is the interpretation of $\bar{g}$?

- If we start with x_1 and **grow $\bar{g}$ each period**, we would end up with x_3 after two periods.

We can also express $\bar{g}$ in terms of the rate of growth of x for each period.

$$\begin{aligned} \cancel{x_1} \cdot (1 + \bar{g})^2 &= x_3 = \cancel{x_1} \cdot (1 + g_1) \cdot (1 + g_2) \Rightarrow \\ (1 + \bar{g})^2 &= (1 + g_1) \cdot (1 + g_2) \Rightarrow \\ \bar{g} &= [(1 + g_1) \cdot (1 + g_2)]^{1/2} - 1 \end{aligned}$$

Average Growth Rates

Let's now **generalize** this result.

Suppose we have a time series on the variable x , represented by: $x_1, x_2, x_3 \dots x_N$, where the subscript refers to time.

The **rate of growth** of x between $t = i$ and $t = i + 1$, g_i , is:

$$1 + g_i = \frac{x_{i+1}}{x_i}$$

The **geometric average rate** of growth of x , $\bar{g}$, is:

$$\bar{g} = \left(\frac{x_N}{x_1} \right)^{\frac{1}{N-1}} - 1$$

and expressed in terms of the rate of growth of x for each period:

$$\bar{g} = \left[\prod_{i=1}^{N-1} (1 + g_i) \right]^{\frac{1}{N-1}} - 1$$